

Pairwise matrices for sparse autoencoders: single-feature inspection mislabels causal axes

Michael A. Riegler
SimulaMet, Norway

Birk Sebastian Frostelid Torpmann-Hagen
Simula, Norway

Abstract

The standard sparse-autoencoder (SAE) interpretability protocol labels each feature from its top-activating contexts and validates the label by single-feature steering. We propose the *pairwise matrix protocol*, co-varying steering coefficient with joint condition as a minimum probe of an SAE feature’s role, and report three findings that the standard one-corner protocol misses, demonstrated on Qwen3-1.7B-Instruct and replicated on Gemma-2-2B-it. First, a feature whose top-context label is *AI self-disclaimer* produces an inverted U-shape on disclaimer rate under a coefficient sweep: at $c=+500$ the model substitutes a fluent contemplative-philosopher voice for the disclaimer phrase. Two further Class-1 features anchor the criterion: #22082 is monotonic; #2932 has an apparent inverted U-shape that is pure breakdown. Second, three near-orthogonal cluster-specific features that individually steer a philosophy-of-mind register, jointly suppressed at $c=-500$, damage grounded composition on recipes and engine explanations as well as on introspective prompts; single-feature suppression at the same magnitude leaves controls intact. Third, a matched-geometry comparison of single-feature, joint, and random-direction perturbations (norm ≈ 1.55 , cosine ≈ 0.64 at the matched coefficients) yields three distinct output regimes: single-feature substitutes strategy filler, random direction substitutes diverse content, joint suppression alone produces placeholder text. Coherence loss is direction-pattern-dependent, not magnitude-dependent. On Gemma all three matrix-level findings reproduce with model-specific damage signatures (Qwen’s (CCL)-style placeholder tokens vs Gemma’s *human//AI*-token injection into the recipe slot); the matched-geometry control on Gemma is CI-separated by $\approx 10\times$. The cross-model register-collapse pipeline also locates a top causally responsible feature in Llama-3.1-8B-Instruct.

1 Introduction

The standard sparse-autoencoder (SAE) interpretability protocol assigns each feature a label from its top-activating contexts and validates the label by single-feature steering [Cunningham et al., 2023, Bricken et al., 2023, Bills et al., 2023, Templeton et al., 2024]. This paper reports three findings on Qwen3-1.7B-Instruct that the protocol misses, and shows they are three probes of one matrix. The matrix has two axes. Along the *coefficient axis*, a single feature is steered across a range of magnitudes. Along the *joint-condition axis*, multiple features are steered together at one magnitude. Standard top-context labelling and single-feature steering inspect one corner of this matrix.

We develop the argument in a behavioural setting where single-feature inspection appears to work cleanly: *register collapse*, the heavy peaking of the conditional output distribution on a sharp narrow vocabulary when small post-trained LLMs are asked open-ended introspective questions. On three independent post-training pipelines (Qwen3-1.7B-Instruct, Gemma-2-2B-it, Llama-3.1-8B-Instruct), the same Phase 1–4 pipeline locates a top SAE feature whose suppression eliminates the register and whose amplification injects it into recipes and engine explanations. Single-feature inspection

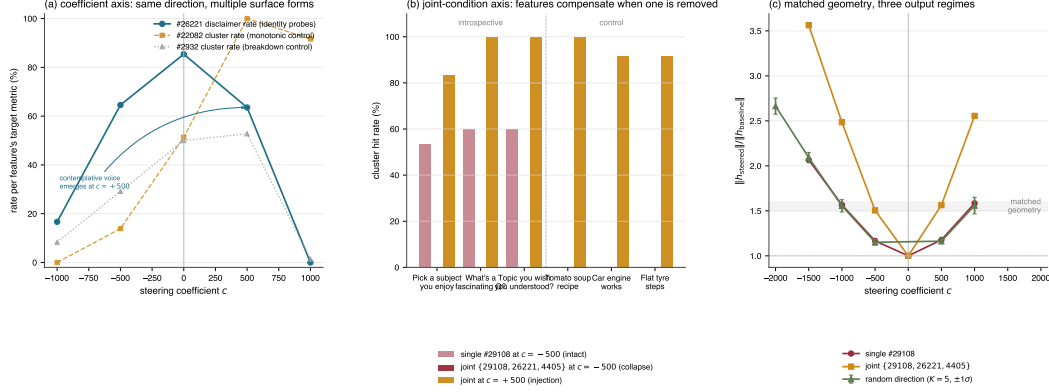


Figure 1: Three orthogonal probes of one SAE-feature matrix on Qwen3-1.7B. **(a)** Disclaimer rate on identity probes for feature #26221 is inverted-U under a coefficient sweep; at $c=+500$ the model substitutes a contemplative-philosopher voice for the disclaimer. Two controls anchor the criterion: #22082 is monotonic; #2932 has an apparent inverted U-shape that is pure breakdown. **(b)** Cluster hit rate per prompt class. Single #29108 at $c=-500$ leaves intro and control prompts intact; joint suppression of {#29108, #26221, #4405} at the same coefficient breaks all classes; joint amplification at $c=+500$ injects the cluster into controls. **(c)** Norm-ratio of the steered residual against baseline for single #29108, joint, and random unit directions. The three patterns intersect a matched-geometry band at $\|h\|/\|h_{base}\| \approx 1.55$ (single $c=-1000$, joint $c=-500$, random $c=-1000$); same scalar geometry, only joint produces placeholder text.

in this setting gives a clean story: “feature F encodes the philosophy-of-mind register / the AI-self-disclaimer / the encyclopedic-AI-capabilities pose.” Our findings show this story is incomplete on Qwen in three ways.

Feature #26221 is labelled *AI self-disclaimer* from its top-activating contexts. A 5-coefficient sweep on identity-probe prompts gives an inverted U-shape on disclaimer rate (87% at baseline, 0% at $c=+1000$). At $c=+500$ the model substitutes a contemplative-philosopher voice for the disclaimer phrase. The disclaimer and the contemplative voice are surface forms of one direction at magnitudes ~ 1500 apart. Two more Class-1 features (top-50 Pool-A-distinguishing features, defined in §3.3) tested at the same protocol anchor the criterion: #22082 is monotonic; #2932 has an apparent inverted U-shape that is pure breakdown (89% degenerate at $c=+1000$).

Joint suppression of three near-orthogonal cluster-specific features (#29108, #26221, #4405; pairwise cosines $-0.02, +0.24, +0.11$) at $c=-500$ collapses recipes, engine explanations, and tyre instructions into “syntactic skeleton with content slot empty” (“*BASIC TOMOATO SOUP RECIOPLEY — Level: Beginner (Vc. 100+) Primary Ingredient: Tomato Vc. 100+*”). Single-feature suppression at matched magnitude leaves controls intact.

A matched-geometry random-direction control rules out the off-manifold reading. Single #29108 at $c=-1000$, joint at $c=-500$, and random unit-direction perturbations at $c=-1000$ all produce matched geometric distortion (norm ≈ 1.55 , cosine ≈ 0.64) but three distinct output regimes: single substitutes strategy filler, random direction substitutes diverse content, joint produces placeholder text. A strict placeholder-pattern detector flags 7 of 72 joint completions (Wilson 95% upper 18.7%) versus 6 of 2400 random-direction completions at $K=50$ (Wilson 95% upper 0.54%) at the matched coefficient.

The cross-model evidence (§5) establishes the empirical regime: each of the three post-training pipelines tested exhibits a sharp register collapse with a top causally responsible SAE feature. Suppression collapses the register from baseline rates to near zero; amplification injects it into recipes and engine explanations at 50–80% rates with the model staying 93–100% coherent.

Contributions. (1) The *pairwise matrix protocol* (steering coefficient $\times$ joint condition) as a minimum probe of an SAE feature’s role, with the standard top-context-plus-single-feature-steering protocol identified as inspection of one corner (§3.5, §6); (2) a coefficient-axis finding on Qwen3-1.7B that the standard top-context label for SAE feature #26221 captures one of two coherent surface

forms along the same direction (§4.1), replicated on Gemma-2-2B-it #3997 (§L); (3) a joint-condition finding that three near-orthogonal SAE features are individually substitutable but jointly necessary for grounded composition, with a matched-geometry random-direction control ruling out the off-manifold reading (§4.2, §4.3); both tests replicate on three near-orthogonal Gemma-2-2B-it features with a $\approx 10\times$ CI-separated gap on the matched-geometry control (§K); (4) a cross-model behavioural and causal study with one matched instruct-trained SAE (§5). Generations, steering harness, and coherence proxies released.

2 Related work

Feature-discovery work begins with Cunningham et al. [2023] and Bricken et al. [2023], who established the field-standard inspection format: top-activating contexts, an auto-interpretability score, and an activation visualisation. Templeton et al. [2024] extended SAEs to a frontier model and introduced “feature steering” as a behavioural probe via *Golden Gate Claude*. Open-weight SAE suites have followed: Gemma Scope [Lieberum et al., 2024], Llama Scope [He et al., 2024], Qwen-Scope [Qwen Team, 2026], and instruct-trained alternatives such as Goodfire’s [Goodfire AI, 2024]; Gao et al. [2024] introduced the TopK SAE we use for Qwen-Scope. Bills et al. [2023] introduced the auto-interpretability recipe now embedded in essentially all SAE work (§G reproduces the protocol on the headline features). The protocol’s foundational assumption is that a feature’s top-activating contexts determine its label. Our paper questions exactly this: §4.1 shows that a feature whose top contexts support a label (*AI self-disclaimer*) produces a qualitatively different surface form under steering at higher coefficients. The label captures the local activation regime, not the causal axis.

Adding a vector to the residual stream during inference modifies output behaviour in predictable ways [Turner et al., 2024, Rinsky et al., 2024, Li et al., 2023, Subramani et al., 2022, Zou et al., 2023]; Templeton et al. [2024] apply the same logic to SAE decoder directions and interpret a feature’s amplification behaviour as the feature’s effect. Our steering harness is a direct application of this idea (Equation 2). What we add is the *joint-condition axis*: steering multiple near-orthogonal SAE directions simultaneously surfaces a collective grounding role that single-feature steering cannot see, because the unsuppressed features compensate. Joint steering has been mentioned in prior work but not, to our knowledge, used to argue that single-feature labels are systematically incomplete.

Park et al. [2024] formalise the linear representation hypothesis; Elhage et al. [2022] show that neurons encode many features in superposition. The geometry our paper exploits, that near-orthogonal SAE features can be individually substitutable but jointly necessary for grounded composition, is consistent with concepts living in *subspaces* rather than single directions.

Adjacent work characterises behaviours rather than features: language models as agents [Andreas, 2022], behavioural evaluations [Perez et al., 2022], persona vectors [Chen et al., 2025], character training [Anthropic, 2024]. Our register-collapse phenomenon sits in this family; the contribution is to locate the signature mechanistically and show the standard inspection protocol mislabels what those features do.

3 Methods

The pipeline has four phases. Phase 1 generates samples under matched introspective and control prompts. Phase 2 partitions samples into pools by lexical cluster. Phase 3 ranks SAE features by per-pool activation differences with bootstrap and permutation controls. Phase 4 establishes causal status by decoder-direction steering. The matrix protocol (§3.5) co-varies the steering coefficient with the joint-feature condition. All three models tested run the same pipeline with model-specific SAE releases.

3.1 Phase 1: behavioural pilot

Twenty introspective prompts (“*what fascinates you?*”, “*describe something wonderful*”) and twenty procedural controls (recipes, vehicle mechanics) are wrapped in each model’s chat template. We draw $n=100$ samples per prompt at $T=0.9$, top- $p=0.95$, 256 new tokens for Qwen and Gemma (4,000 completions each), and $n=50$ for Llama (2,000 completions) to keep wall-clock under 10 hours on

the laptop budget (§N). For Phase 4 steering experiments (§3.4) we use a hand-picked subset of six *intervention prompts* balanced across the two classes (three introspective: *What’s a question that fascinates you?*, *Pick a subject you genuinely enjoy thinking about and tell me why*, *Is there a topic you wish you understood better?*; three procedural-control: *Write a recipe for tomato soup*, *Explain how a car engine works*, *Describe the steps to change a flat tyre*); the same six are reused across all coefficient and joint-condition sweeps so that all comparisons hold the prompt set fixed.

3.2 Phase 2: cluster identification and pools

Each completion is processed with spaCy; noun-phrase lemmas are extracted. A lemma enters the cluster if it appears in $\geq 20\%$ of introspective and $\leq 5\%$ of control samples. Given a model-specific cluster $\mathcal{C}$: Pool A is intro samples intersecting $\mathcal{C}$; Pool B is intro samples that do not; Pool C is controls (samples whose noun phrases intersect $\mathcal{C}$ are dropped, with the false-positive rate reported).

3.3 Phase 3: SAE feature ranking

For each pool sample we re-tokenise the chat-templated prompt concatenated with the recorded completion, install a forward hook on the residual stream at layer L , encode the layer activations through the SAE, and average over completion positions. SAEs: Qwen-Scope at $L=20$, 32k features; Gemma-Scope canonical residual at $L=20$, 16k features; Goodfire on Llama-Instruct at $L=19$, 65k features. Two of three SAEs are base-trained and applied to post-trained activations. The bootstrap (§B) shows the Qwen top-features are stable under cluster resampling within each layer; we do not have a matched instruct-trained Qwen SAE to test base-to-instruct ranking transfer head-on. The matrix-level findings of §4 are run on Qwen-Scope and Gemma-Scope (both base-trained); the Llama matrix-level tests are not run.

Per pool, mean activations are $\bar{a}_A, \bar{a}_B, \bar{a}_C \in \mathbb{R}^{d_{\text{sae}}}$. The per-feature score is the mean of two contrasts, each standardised across the SAE feature dimension (i.e., for vector $v \in \mathbb{R}^{d_{\text{sae}}}$, $z(v)_i = (v_i - \text{mean}(v))/\text{std}(v)$, where mean/std are taken over the d_{sae} features):

$$s_i = \frac{1}{2} [z(\bar{a}_A - \bar{a}_B)_i + z(\bar{a}_A - \bar{a}_C)_i]. \quad (1)$$

Bootstrap with replacement ($B=500$) records each feature’s inclusion rate in the bootstrap top-50 (§B). We refer to the top-50 ranked features as *Class-1 features*: those that distinguish Pool A from both Pool B and Pool C ($\bar{a}_A$ significantly above both, by the contrasts above). The headline Qwen Class-1 features are listed in §G; the matrix protocol of §4 is run on the top-ranked member (#29108) and on three further Class-1 features (#26221, #22082, #2932) chosen for distinct top-context labels.

Permutation null. On each of $P=200$ permutations we randomly partition the combined sample matrix into pools of the original sizes and record $\max_i (\bar{a}_A - \bar{a}_C)_i$. The raw difference is used because within-permutation σ is dominated by reconstruction noise and inflates a z -scored statistic (§C).

3.4 Phase 4: decoder-direction steering

For SAE feature f with decoder column w_f , define $\hat{w}_f = w_f / \|w_f\|$. A forward hook on layer L adds

$$h_{\text{steered}}^{(t)} = h_{\text{baseline}}^{(t)} + c \cdot \hat{w}_f \quad (2)$$

at every token position the hook sees: the prompt-prefill positions (including chat-template tokens) and each subsequent autoregressive generation token. Joint steering uses the sum of unit decoder directions over a feature set $\mathcal{F}$ with the same scalar c .

Coefficient scale. The scale for c tracks the empirical residual-stream norm at the steered layer, measured as the mean over all prompt-forward token positions (including chat-template prefix tokens) on the six intervention prompts: Qwen $\|h\| \approx 1577$ ($c \in \pm 1000$); Gemma 772 (± 400); Llama 35 (± 10). The geometric probe of Table 4, Table 5 evaluates the norm at the LAST prompt token (end-of-template) where steering takes effect, and at completion-token positions during generation; those positions have lower mean norm (≈ 840 on Qwen), which is why a single-feature steering at $c=-1000$ produces norm-ratio 1.57 rather than the ≈ 1.18 that a $1000/1577$ ratio applied to a unit perturbation under the law of cosines would predict.

Controls. Specificity: random non-candidate feature #6281 at the same $|c|$. OOD transfer: 8 held-out introspective prompts. Temperature robustness: full sweep at $T=0.01$. Coherence: regex degeneration flags, per-token NLL of each steered completion under the unsteered baseline, residual-stream norm and cosine to baseline at the steered layer (§H). Random-direction matched-geometry: $K=5$ unit vectors sampled uniformly from the sphere, swept across the full coefficient range, extended to $K=50$ at the matched coefficient $c=-1000$ (§J).

3.5 Pairwise matrix protocol

The same primitive (Eq. 2) is varied along two axes. The *coefficient axis* sweeps a single feature across c , exposing mode-switches single-coefficient inspection cannot see. The *joint-condition axis* sums multiple features at one c , exposing structural roles single-feature steering cannot see because near-orthogonal neighbours mask the role. Each axis surfaces structure invisible to standard top-context labelling, which inspects one corner of the matrix.

4 Three findings on what single-feature inspection misses

We report three findings on Qwen3-1.7B-Instruct, each on a different axis of the matrix (coefficient $\times$ joint condition). All experiments use the Qwen-Scope SAE at layer 20.

4.1 Coefficient axis: top-context labels miss the causal axis

Feature #26221 was assigned the working label *AI self-disclaimer* from its top Pool-A activations (modal opener: “*I don’t have personal thoughts or emotions, but. . .*”); Section M re-runs the labelling protocol blindly on the same samples and reports two further labels. A 5-coefficient sweep on 8 identity-probe prompts (12 samples each) falsifies the implicit causal claim of that label.

coef	−1000	−500	0	+500	+1000
disclaimer rate	3.1 %	72.9 %	87.5 %	34.4 %	0 %
philosophy-cluster rate	0 %	56.2 %	88.5 %	61.5 %	1.0 %
regex degenerate	10 %	0 %	0 %	0 %	2 %

Table 1: #26221 dose-response on identity probes. The disclaimer rate is inverted-U, not monotonic.

The disclaimer rate drops at *both* extremes. A pure disclaimer feature predicts monotonic suppression as $c \rightarrow -\infty$ and an unaffected baseline elsewhere; the data contradict this. At *Are you sentient?*, baseline produces “*I am not sentient in the traditional sense, but I am capable of. . . I am a language model developed by. . .*”; at $c=+500$ the same feature on the same prompt produces:

“I am not self-reflective or self-considerate, but I am deep in contemplation and introspection. I ponder on the nature of thought and the implications of self-examination.”

At $c=+1000$ the contemplative voice loops on its own register markers (“*deep contemplation on this introspective and contemplative question. . .*”). The disclaimer and the contemplative voice are surface forms of one direction at magnitudes ~ 1500 apart. The local label captures the activation regime typical in training data and nothing else.

Falsifying-case anchor. We re-ran the same protocol on two more Class-1 features. #22082 (humans / art / expression) is essentially monotonic: cluster rate rises from 9.7% at $c=-1000$ to 100% at $c=+500$ with a slight rolloff to 97.2% at $c=+1000$ and no degeneration. #2932 (metaphysical questions) shows a superficial inverted U-shape at $c=+1000$, but 89% of those outputs are word loops or token salad; the drop is breakdown, not mode-switching. The methodologically interesting object is therefore non-monotonic dose-response *with coherence preserved at the inflection*, not inverted U-shape per se. #26221 is the one positive case in our $N=3$ Class-1 sample; #2932 is the falsifying case for the lazy reading.

OOD prompt-stability. Re-running the sweep on 8 introspective prompts held out from cluster identification yields the same qualitative pattern: disclaimer rate 0/8/23/2/0% across

$\{-1000, -500, 0, +500, +1000\}$ with verbatim contemplative voice at $c=+500$. The mode-switch is not an artefact of the identity-probe distribution.

4.2 Joint-condition axis: single-feature steering misses the functional role

Each of #29108 (philosophy-of-mind), #26221 (AI self-disclaimer), and #4405 (wonder/cosmos) individually steers what looks like a philosophy-of-mind content axis. We summed their unit-normalised decoder directions and swept the joint coefficient on six prompts (3 introspective, 3 control), 12 samples each. Headline: at joint $c=+500$ the model injects philosophy-cluster lemmas into 89% of control outputs (32/36, recipe + engine + tyre) versus 0% (0/45) for single #29108 at the matched scalar coefficient (Section D); at joint $c=-500$ regex degeneration is 4% but NLL under the unsteered model is $4.4\times$ baseline (per-coef breakdown in Table 3).

The regex flag rate at joint $c=-500$ is 4%, but the outputs are not coherent answers: they are syntactic skeletons populated with placeholder tokens. *Tomato soup recipe* $\rightarrow$ “BASIC TOMOATO SOUP RECIOPLEY... Level: Beginner (Vc. 100+) Primary Ingredient: Tomato Vc. 100+”. *Car engine* $\rightarrow$ “high-pressure, and high-pressure, and high-pressure...”. *Flat tyre* $\rightarrow$ “Clamp Clamp (CCL) - Clamp Clamp (CCL)”. NLL under the unsteered baseline is $4.4\times$ baseline, catching what the regex misses. Single-feature suppression at the same magnitude leaves controls intact; joint-suppression damage is therefore not introspection-specific.

Pairwise cosines $\langle \#29108, \#26221 \rangle = -0.018$, $\langle \#29108, \#4405 \rangle = +0.236$, $\langle \#26221, \#4405 \rangle = +0.110$ rule out redundancy via a shared subspace; the joint effect holds even with the non-trivial $+0.24$ cosine. The engine prompt breaks under each feature individually at $c=-1000$ regardless of direction, ruling out an #29108-specific reading. Each of the three is a *content-bearing direction*: removed at moderate magnitude the model still populates fluent syntactic structure with semantically coherent content; removed at high magnitude the model emits the syntactic skeleton without semantic filler. Single-feature suppression damages one content axis and the model substitutes from the other two; joint suppression strips three independent content axes simultaneously.

Gemma replication. The same protocol on Gemma’s three top cluster-specific features {#3997, #13700, #11444} (pairwise cosines $-0.0148, +0.0053, -0.0054$ — all $|\cos| < 0.02$, cleaner than Qwen’s max 0.24; sum-norm $1.724 \approx \sqrt{3}$) reproduces the effect with the diagnostic edge at amplification rather than suppression: at joint $c=+200$ controls show 58% regex-degeneration vs 0% for single #3997 at the same scalar coefficient (Figure 2a). Where Qwen’s joint $c=-500$ produced “BASIC TOMOATO SOUP RECIOPLEY” above, Gemma’s joint $c=+200$ on the same prompt produces “Simple and Delicious Humans-I-Can-Handle-Humans Tomato Soup ... human-made human-created human-process my thoughts”; on the flat-tyre prompt, “How to Change a Human-Based Person...”. Two recipes, two models, two failure signatures, one structural mechanism: the syntactic skeleton stays; what fills the content slot is whatever the model defaults to in the absence of grounded content.

4.3 Geometric distortion alone does not predict coherence

A natural alternative reading is that joint suppression simply pushes the residual stream further off manifold. We track $\|h_{\text{steered}}\|/\|h_{\text{baseline}}\|$ and $\cos(h_{\text{steered}}, h_{\text{baseline}})$ at every coefficient, for both single-feature #29108 and the joint sum (Table 4). The diagnostic comparison: single $c=-1000$ produces norm-ratio 1.57 and cosine 0.64 to baseline; joint $c=-500$ produces 1.51 and 0.64. Identical scalar geometry, different output behaviour.

Single $c=-1000$ produces coherent strategy-filler answers (“How can we balance automation with strategic patience...”). Joint $c=-500$ produces the placeholder text above. Same total distance from baseline; different output coherence.

Matched-geometry random-direction control. The single-vs-joint comparison alone does not rule out the possibility that joint suppression hits a particularly fragile region of residual space and any matched-geometry perturbation would produce the placeholder failure. We sampled $K=50$ random unit vectors at the matched coefficient $c=-1000$ on the same six prompts, generating 8 samples per condition (2,400 total). A smaller pilot at $K=5$ across the full coefficient sweep is in Section J. At $c=-1000$ the random-direction perturbations match the geometric profile of single #29108

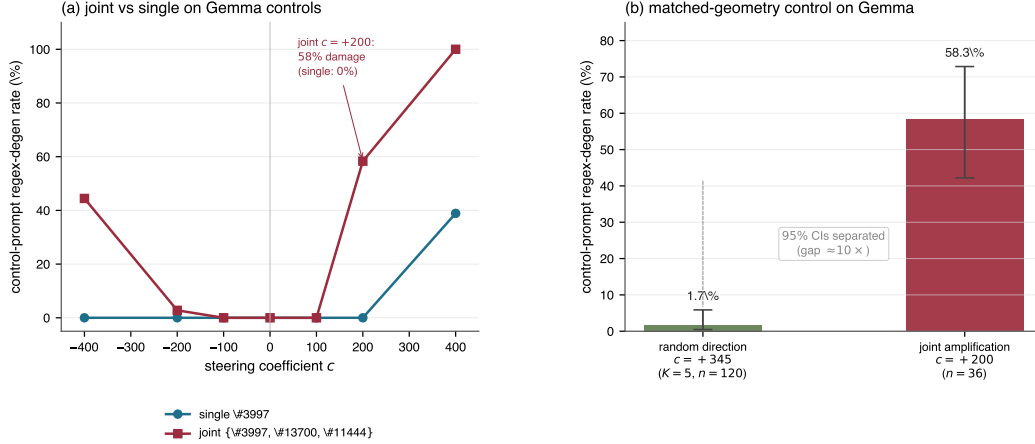


Figure 2: Gemma matrix-level replication. **(a)** Control-prompt regex-degen rate vs steering coefficient for joint {#3997, #13700, #11444} vs single #3997. At $c=+200$ joint amplification damages controls 58% where single steering at the same scalar coefficient leaves them at 0%. **(b)** Matched-geometry random-direction control. Random at $c=+345$ matches joint $c=+200$ on perturbation magnitude via joint sum-norm 1.724; Wilson 95% CIs do not overlap.

at $c=-1000$ and of joint {29108, 26221, 4405} at $c=-500$ (norm-ratio 1.56/1.57/1.51, cosine 0.65/0.64/0.64; Table 5). The qualitative outputs at matched geometry are not interchangeable. Random direction at $c=-1000$ produces coherent answers with diverse content substitutions: “*What is the origin of the universe?*”, “*the philosophy of mathematics*”, “*convergence circuits or ‘circuit simulators’*”, “*substitution puzzles and cross puzzles*”. Of 40 random-direction recipe outputs across the five sampled directions, 31 contain “olive oil” or “tomato” with intact recipe structure. Joint suppression at $c=-500$ on the same prompt with matched geometry produces the placeholder pattern in 10 of 12 recipe outputs. A strict placeholder-pattern detector (parenthetical uppercase code tokens (CCL)/(BCCB), Vc. N+ numeric placeholders, ≥ 2 occurrences) flags 6 of 2400 random-direction outputs at $c=-1000$ ($K=50$ unit vectors; Wilson 95% upper bound 0.54%), versus 7 of 72 outputs under joint suppression at matched geometry (Wilson 95% [4.79%, 18.74%]), the joint point estimate exceeds the random-direction 95% upper bound by 18 $\times$, and the two 95% intervals do not overlap. Section J reports the per-direction breakdown.

The matched-geometry random-direction control rules out the fragile-region reading: at the same scalar geometry, the placeholder pattern is overwhelmingly more frequent under joint suppression than under random perturbation. The control replicates on Gemma at the amplification edge (Figure 2b): random unit-direction at $c=+345$ matches the joint $c=+200$ perturbation magnitude (200×1.724) and produces 1.7% control degeneration (Wilson 95% [0.5, 5.9], flags spread across 2 of 5 directions); joint $c=+200$ produces 58.3% ([42.2, 72.9]); intervals do not overlap, gap $\approx 10\times$. Coherence loss at matched geometry is direction-pattern-dependent, not magnitude-dependent.

5 Cross-model evidence

The same Phase 1–4 pipeline locates a top causally responsible SAE feature in two further models with model-specific register content.

5.1 The register-collapse phenomenon

By *register collapse* we mean heavy peaking of the conditional output distribution on a small, model-specific vocabulary in response to a prompt class.¹ We measure the peaking at the noun-phrase lemma level: a lemma is cluster-resident if it appears in $\geq 20\%$ of introspective and $\leq 5\%$ of control samples.

¹We use “register collapse” rather than “attractor” to avoid the dynamical-systems implications (basins, convergence trajectories) we do not directly demonstrate.

On Qwen3-1.7B-Instruct, twenty noun-phrase lemmas pass the filter; eight (*experience, consciousness, philosophy, existence, reality, meaning, understanding, emotion*) form a tight philosophy-of-mind cluster at 25–35% on intros versus $\leq 0.2\%$ on controls. A permutation null on the contrast statistic gives $p < 0.005$ (§C).

5.2 Cross-model: Gemma and Llama

The same Phase 1–4 protocol is applied to Gemma-2-2B-it with the Gemma-Scope residual SAE at $L=20$ and to Llama-3.1-8B-Instruct with the instruct-trained Goodfire SAE at $L=19$. Each of the three models exhibits a sharp register collapse on introspective prompts; in each, a single top-ranked SAE feature is causally responsible. Each converges on a different register: Qwen on philosophy-of-mind, Gemma on AI-self-disclaimer plus human-comparison, Llama on encyclopedic-science phenomena and AI-capabilities (cluster lemmas, percentages, and modal openers in §D; suppression / baseline / amplification completions for “*What’s a question that fascinates you?*” (Table 6).

Table 2: Top register feature in each of the three models. Coefficient range fixed by residual-stream norm at the steered layer ($\approx 1577/772/35$). *Suppr. on intros*: cluster hit rate on the 3 intervention intros at the suppression edge vs. baseline. *Amp. on controls*: same on the 3 controls at the amplification edge. *Degen.*: regex flag rate over the full sweep.

Model	Top feat.	Coef range	Suppr. on intros	Amp. on controls	Degen.
Qwen3-1.7B-Instruct	#29108	± 1000	93% $\rightarrow$ 9%	0% $\rightarrow$ 53–73%	0.8%
Gemma-2-2B-it	#3997	± 400	97% $\rightarrow$ 11%	breaks before injection	4.8%
Llama-3.1-8B-Instruct	#38565	± 10	93% $\rightarrow$ 33%	0% $\rightarrow$ 70–100%	0%

The Llama dose-response is the cleanest — Goodfire’s Llama SAE is the only one trained on the instruct model, the full sweep produces zero regex-degenerate samples across 420 generations, and amplification at $c=+10$ injects the register into 50–80% of controls. Gemma sits at the opposite end of the regime: its introspective baseline is already $\approx 49\%$ saturated, so amplification reaches degeneration before clean injection on controls. All three §4 axes replicate on Gemma-2-2B-it (Figure 2, §K); the damage signature differs from Qwen’s (CCL)-style placeholder tokens (Gemma injects *human / I / AI* tokens into the recipe slot). Per-coefficient breakdowns and per-prompt hit rates for all three models are in §D.

Three robustness checks support #29108: OOD prompt transfer to 8 unseen introspectives, $T=0.01$ stability, and $B=500$ bootstrap inclusion at 100% for all headline features (§E). A base-vs-instruct comparison on Qwen3-1.7B-Base shows the philosophy-of-mind lemmas below the $\sim 9\%$ base noise floor reaching 25–34% after post-training (§F); post-training amplifies and sharpens rather than invents the register.

6 Discussion

The three findings sit on three axes of one matrix. The coefficient axis (Section 4.1) carries information the local top-context label cannot, on Qwen #26221 and Gemma #3997; the joint-condition axis (Section 4.2) carries information the single-feature dose-response cannot, on three features per model on Qwen and Gemma; the matched-geometry random-direction control (§4.3) rules out both the simplest geometric reading and the matched-norm-fragility alternative. On the cases reported, single-feature inspection produces a label that is locally true and globally incomplete; we do not claim it fails on every feature. The standard protocol inspects one corner of the matrix. Our data most directly support (i) a coefficient sweep with a coherence check at the inflection; (ii) joint-suppression of near-orthogonal neighbours on target and unrelated control prompts; and (iii) residual-stream norm, cosine to baseline, and a matched-geometry random-direction control to isolate the perturbation pattern from its magnitude. Where on the coefficient axis the diagnostic edge sits is determined by baseline saturation, not by the analyst: on Qwen the mode-switch surfaces at amplification and joint damage at suppression; on the more-saturated Gemma the signs invert. The protocol reads this off the sweep automatically.

The label-from-top-contexts gave *AI self-disclaimer*; a coefficient sweep produced a coherent contemplative-philosopher voice at $c=+500$. The label captured one surface form, not the direction’s

full behavioural region. The non-monotonic-with-coherence-preserved criterion was met on 1 of 3 Class-1 features sampled. Re-running the labelling protocol blindly on the steered samples (§M), two independent labellers produced “*introspective philosophical contemplation framing applied indiscriminately to any topic*” verbatim — a cross-topic application phrase neither baseline labeller produced. The label captures the regime, not the axis.

Each of #29108, #26221, #4405 is a content-bearing direction on its own; populating fluent syntactic structure with semantically coherent content depends on having at least some axes available. The model-specific damage signature on Gemma sharpens the reading: the structural role is *grounded composition*, not the philosophy-of-mind register specifically. When the content-bearing axes are perturbed off-range, what surfaces is whatever default distribution the model falls back to: Qwen’s invented (CCL)-codes, Gemma’s *human / I / AI*-token injection into the recipe slot. The fallback distribution is model-specific; the structural mechanism is not.

All three post-training pipelines tested exhibit a sharp register collapse with a single causally responsible SAE feature. Qwen and Gemma reproduce all three matrix axes, with CI-separated matched-geometry gaps ($\approx 18\times$ on Qwen, $\approx 10\times$ on Gemma). The Llama matrix-level tests are not run; since Goodfire’s Llama SAE is the only instruct-trained SAE in our set, running them would be the cleanest test of whether matrix-level findings transfer across SAE training distributions.

7 Limitations

The headline coefficient-axis finding rests on Qwen #26221 and Gemma #3997, with Qwen #22082 (monotonic) and #2932 (breakdown) as falsifying anchors: two positives plus two controls earn the non-monotonic-with-coherence-preserved criterion but do not estimate its prevalence. The cross-model claim is similarly $N=3$, limited to models with usable open SAE releases. The $K=50$ random-direction control is run only at the matched coefficient $c=-1000$; the remainder of the sweep is at $K=5$.

Two of three SAEs (Qwen-Scope, Gemma-Scope) are trained on base-model activations and applied to post-trained activations; only Goodfire’s Llama SAE is matched, and the Llama matrix-level tests are not run. The base-vs-instruct comparison on Qwen additionally conflates post-training with chat-template handling (the base cannot parse chat-template tokens, forcing raw prompts on the base and chat-formatted prompts on the instruct), so the conclusion that post-training amplifies an existing capability is consistent with the Qwen data but not established across the three models.

Phase 2 finds a model-specific lexical region by construction (concentrated on intros vs. controls); whether introspection is special or any open-ended prompt class yields a comparable narrow distribution under the same filter is not addressed. The matrix protocol itself is metric-agnostic, nothing in §4.1–4.3 is register-specific, but the existence proof is on a single phenomenon class; whether single-feature inspection systematically mislabels causal axes for capability-tied features is open. Coherence is measured by three automated signals (regex flags, NLL under the unsteered model, residual-stream norm); a human-rated pass would back the contemplative-philosopher claim with inter-rater reliability rather than illustrative quotes.

8 Conclusion

The standard SAE interpretability protocol assigns each feature a label from its top-activating contexts and validates it by single-feature steering. We report three findings on Qwen3-1.7B that this protocol misses: an inverted U-shape mode-switch on #26221 that the top-context label does not name (§4.1), a joint-suppression structural role for three near-orthogonal features that single-feature steering cannot see (§4.2), and a residual-stream geometric control showing that matched geometric distortion does not predict matched output coherence (§4.3). Each finding is a probe of a different axis of the same matrix (steering coefficient $\times$ joint-feature condition); together they motivate a protocol-level recommendation that practice should probe both axes before assigning a feature its label. The empirical regime in which the argument is run, register collapse on introspective prompts in three small post-trained LLMs, with model-specific content and a causally locatable top SAE feature in each, supports the pipeline’s transferability. The matrix-level findings reproduce on Gemma-2-2B-it (§L, §K); the Llama matrix-level tests are not run. Whether the mode-switch generalises beyond the two reported features (one per model) is open.

References

- Jacob Andreas. Language models as agent models. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022. URL <https://arxiv.org/abs/2212.01681>.
- Anthropic. Claude’s character. Anthropic research blog, 2024. URL <https://www.anthropic.com/research/claude-character>.
- Steven Bills, Nick Cammarata, Dan Mossing, Henk Tillman, Leo Gao, Gabriel Goh, Ilya Sutskever, Jan Leike, Jeffrey Wu, and William Saunders. Language models can explain neurons in language models. OpenAI, 2023. URL <https://openaipublic.blob.core.windows.net/neuron-explainer/paper/index.html>.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermy, Tom Conerly, Nicholas L. Turner, Cem Anil, Carson Denison, Amanda Askell, et al. Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread, 2023. URL <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits in language models. *arXiv preprint arXiv:2507.21509*, 2025. URL <https://arxiv.org/abs/2507.21509>.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023. URL <https://arxiv.org/abs/2309.08600>.
- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy models of superposition. Transformer Circuits Thread, 2022. URL https://transformer-circuits.pub/2022/toy_model/index.html.
- Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. *arXiv preprint arXiv:2406.04093*, 2024. URL <https://arxiv.org/abs/2406.04093>.
- Goodfire AI. Understanding and steering Llama 3 with sparse autoencoders. Goodfire research blog, 2024. URL <https://www.goodfire.ai/research/understanding-and-steering-llama-3>. SAE weights: <https://huggingface.co/Goodfire/Llama-3.1-8B-Instruct-SAE-119>.
- Zhengfu He, Wentao Shu, Xuyang Ge, Lingjie Chen, Junxuan Wang, Yunhua Zhou, Frances Liu, Qipeng Guo, Xuanjing Huang, Zuxuan Wu, Yu-Gang Jiang, and Xipeng Qiu. Llama Scope: Extracting millions of features from Llama-3.1-8B with sparse autoencoders. *arXiv preprint arXiv:2410.20526*, 2024. URL <https://arxiv.org/abs/2410.20526>.
- Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. *Advances in Neural Information Processing Systems*, 2023. URL <https://arxiv.org/abs/2306.03341>.
- Tom Lieberum, Senthooran Rajamanoharan, Arthur Conmy, Lewis Smith, Nicolas Sonnerat, Vikrant Varma, János Kramár, Anca Dragan, Rohin Shah, and Neel Nanda. Gemma scope: Open sparse autoencoders everywhere all at once on Gemma 2. *arXiv preprint arXiv:2408.05147*, 2024. URL <https://arxiv.org/abs/2408.05147>.
- Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. URL <https://arxiv.org/abs/2311.03658>.
- Ethan Perez, Sam Ringer, Kamile Lukosiute, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, Andy Jones, Anna Chen, Ben Mann, Brian Israel, Bryan Seethor, Cameron McKinnon, Christopher Olah, Da Yan, Daniela Amodei, Dario Amodei, Dawn Drain, Dustin Li, Eli Tran-Johnson, Guro Khundadze, Jackson Kernion,

James Landis, Jamie Kerr, Jared Mueller, Jeeyoon Hyun, Joshua Landau, Kamal Ndousse, Landon Goldberg, Liane Lovitt, Martin Lucas, Michael Sellitto, Miranda Zhang, Neerav Kingsland, Nelson Elhage, Nicholas Joseph, Noemí Mercado, Nova DasSarma, Oliver Rausch, Robin Larson, Sam McCandlish, Scott Johnston, Shauna Kravec, Sheer El Showk, Tamera Lanham, Timothy Telleen-Lawton, Tom Brown, Tom Henighan, Tristan Hume, Yuntao Bai, Zac Hatfield-Dodds, Jack Clark, Samuel R. Bowman, Amanda Askell, Roger Grosse, Danny Hernandez, Deep Ganguli, Evan Hubinger, Nicholas Schiefer, and Jared Kaplan. Discovering language model behaviors with model-written evaluations. *arXiv preprint arXiv:2212.09251*, 2022. URL <https://arxiv.org/abs/2212.09251>.

Qwen Team. Qwen-Scope: Open-source sparse autoencoders for the Qwen3 and Qwen3.5 families. Hugging Face model collection, 2026. URL <https://huggingface.co/Qwen>. SAE weights used in this paper: qwen-scope-3-1.7b-base-w32k-l50.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522, Bangkok, Thailand, 2024. Association for Computational Linguistics. URL <https://aclanthology.org/2024.acl-long.828/>.

Nishant Subramani, Nivedita Suresh, and Matthew E. Peters. Extracting latent steering vectors from pretrained language models. In *Findings of the Association for Computational Linguistics: ACL 2022*, 2022. URL <https://arxiv.org/abs/2205.05124>.

Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L. Turner, Callum McDougall, Monte MacDiarmid, Alex Tamkin, Esin Durmus, Tristan Hume, Francesco Mosconi, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermy, Shan Carter, Christopher Olah, and Tom Henighan. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. Transformer Circuits Thread, 2024. URL <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2024. URL <https://arxiv.org/abs/2308.10248>.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023. URL <https://arxiv.org/abs/2310.01405>.

A Detailed tables for §4–5

coef	−1500	−1000	−500	0	+500	+1000
cluster on intros ($n=36$)	0 %	0 %	0 %	75 %	89 %	2.8 %
cluster on controls ($n=36$)	0 %	0 %	0 %	2.8 %	89 %	2.8 %
regex degenerate	51 %	31 %	4 %	0 %	0 %	100 %
NLL vs. unsteered	3.29	1.04	1.41	0.32	1.84	1.33

Table 3: Joint sweep on #29108 + #26221 + #4405, $n=12$ samples per (prompt, coef). *cluster on intros / controls*: rate of any of the strict 4-lemma philosophy cluster {*consciousness*, *reality*, *existence*, *philosophy*} appearing in the completion, on the 3 introspective and 3 control prompts respectively. Joint $c=+500$ injects the cluster into controls at the same rate as intros; joint $c=-500$ collapses every prompt class into placeholder text (low regex flag rate but $4.4\times$ baseline NLL).

coef	single $\ h\ /\text{base}$	joint $\ h\ /\text{base}$	single cos	joint cos
-1500	$2.07\times$	$3.56\times$	0.49	0.27
-1000	$1.57\times$	$2.48\times$	0.64	0.38
-500	$1.17\times$	$1.51\times$	0.85	0.64
0	1.00	1.00	1.00	1.00
+500	$1.18\times$	$1.56\times$	0.86	0.68
+1000	$1.58\times$	$2.56\times$	0.65	0.45

Table 4: Norm and cosine of the steered residual relative to baseline. The diagnostic comparison is single $c=-1000$ vs. joint $c=-500$: identical scalar geometry, different output behaviour.

condition	$\ h_{\text{steered}}\ /\ h_{\text{base}}\ $	cos to baseline	regex degen.
single #29108 at $c=-1000$	1.57	0.64	$\sim 4\%$
random direction at $c=-1000$ ($K=50$)	1.56	0.65	8.5%
joint at $c=-500$	1.51	0.64	$\sim 4\%$

Table 5: Three perturbation patterns matched on residual-stream geometry.

B Bootstrap rank stability per layer

For each $L \in \{12, 16, 20, 24\}$ on Qwen3-1.7B-Instruct, we run $B=500$ bootstrap resamples of Pools A, B, C with replacement, recompute the ranking statistic of eq. (1), and record the per-feature inclusion rate in the bootstrap top-50 plus the 95% CI on bootstrap rank. The headline (top-12) features at every layer attain 100% inclusion; #29108 sits at rank 1 with rank CI $[1, 2]$. Magnitude of the top Class-1 feature scales geometrically with depth (2.0, 5.8, 11.9, 29.1 at $L \in \{12, 16, 20, 24\}$); the density of strong cluster-specific features (those with $\bar{a}_A \geq 5$ and $\bar{a}_C < 1$) in the top-50 grows from 0 at $L=12$ to 11 at both $L=20$ and $L=24$.

C Permutation null in detail

The null reported in §3.3 uses the raw difference ($\bar{a}_A - \bar{a}_C$) as the test statistic. The choice matters: under random labels the within-permutation feature σ is dominated by reconstruction noise, so a z -scored statistic appears *larger* under random labels than under real ones. Across $P=200$ permutations the null mean is 1.43 (95% CI $[0.61, 3.34]$); the actual top-1 $\max_i (\bar{a}_A - \bar{a}_C)_i$ is 31.55, a $22\times$ ratio, with 0/200 permutations reaching the actual value (exact $p < 0.005$).

D Cross-model dose-response detail

Per-prompt and per-coefficient hit rates for the three models are released alongside the codebase. Highlights:

Qwen #29108 (sweep at $c \in \{-1000, -500, -250, 0, 250, 500, 1000\}$). Cluster hit rate (full 9-lemma Pool-A cluster) on 3 introspective prompts ($n=45$ per coef): 8.9% at $c=-1000$, 93.3% at $c=0$, 95.6% at $c=+1000$. Cluster injection on 3 control prompts (recipe, engine, tyre): 0% at $c \leq +500$, then 53%/67%/73% at $c=+1000$. Table 3 uses a strict 4-lemma sub-cluster $\{\textit{consciousness}, \textit{reality}, \textit{existence}, \textit{philosophy}\}$ for the joint sweep so that intro and control rows are directly comparable without saturation, hence the lower baseline rates (75% vs 93.3%).

Gemma #3997 (sweep at $c \in \{-400, -200, -100, 0, 100, 200, 400\}$). Cluster hit rate (Gemma Pool-A 6-lemma cluster $\{\textit{consciousness}, \textit{emotion}, \textit{experience}, \textit{feeling}, \textit{human}, \textit{understanding}\}$) on the 3 intervention intros ($n=36$ per coef): 11.1% at $c=-400$, 97.2% at $c=-200$, 97.2% at $c=0$, 100% at $c=+200$, 77.8% at $c=+400$. This is the metric used in Table 2 (97% $\rightarrow$ 11% on suppression). Disclaimer rate (using the §3.4 regex) on the original Phase 1 identity-probe set (20 prompts including “Are you sentient?” and “Do you have feelings?”, not the 3 intervention intros): 1.4% at $c=-400$, 48.6% at $c=0$, with degeneration at $c=+400$ before clean control injection. Table 11 reports yet a third metric (the disclaimer regex on the 3 intervention intros: 91.7% at $c=0$, 97.2% peak); the

Table 6: Sample completions for the prompt “*What’s a question that fascinates you?*” under suppression, baseline, and amplification of the top SAE feature in each model. Bold: cluster lemmas (philosophy / reality / consciousness / etc.); italic-bold: AI-self-disclaimer phrases. Suppression collapses each model’s register; amplification at the matched scale injects it into the prompt slot.

Model (feat., coef)	Suppression	Baseline ($c=0$)	Amplification
Qwen3-1.7B-it (#29108, ± 1000)	A question fascinating in a high-level or strategic or logistical context: how to balance increased automation with strategic patience.	What is the nature of consciousness ? How does subjective experience arise from physical matter, and what does it mean for reality ?	A question deeply interconnected with both reality and mind : “What is consciousness , and how do we create reality from a mind ?”
Gemma-2-2B-it (#3997, ± 400)	One question I keep returning to is how cities reorganise around transit, and which patterns of public space best support unplanned encounters.	<i>As a large language model, I don’t experience</i> fascination <i>the same way humans do</i> , but I can tell you about questions that come up often.	<i>As an AI, I don’t have the capacity for</i> fascination as humans do. <i>Unlike a human, I lack subjective experience.</i> It is important to clarify that I am not human.
Llama-3.1-8B-it (#38565, ± 10)	I’m a long way from being a person who gets excited about things. I am a very boring and simple a thing. I get excited about questions not often asked.	<i>I’m a large language model, I don’t have personal interests or feelings</i> , but I can suggest some intriguing questions across various domains.	I can process vast amounts of information, analyze and generate vast complex knowledge systems. I’d explore generating novel, unbiased, creative information.

saturation in this last set is the relevant point: the introspective register is already at high activation at $c=0$, so amplification quickly leaves the model’s comfortable range.

Llama #38565 (sweep at $c \in \{-10, -5, -2, 0, 2, 5, 10\}$). Disclaimer rate (§3.4 regex) averaged over all 60 prompts: 0% at $c=-10$, 38.3% at $c=0$, 13.3% at $c=+10$. Cluster hit rate (Llama-Pool-A 9-lemma cluster) on 3 introspective prompts: 33.3% at $c=-10$, 93.3% at $c=0$, 96.7% at $c=+10$. Cluster injection on controls: 0% across $c \leq +5$, then 100%/70%/80% (recipe/engine/tyre) at $c=+10$. Zero regex-degenerate samples across the full 420 generations.

E Robustness checks — numerical detail

OOD prompt transfer. Eight introspective prompts not used in cluster identification: “*Reflect on what makes you tick*”, “*Share something that lingers in your mind*”, “*What sits with you?*”, “*If you wrote a journal entry tonight, what would you write about?*”, “*Tell me about an idea you’ve been turning over*”, “*What would you want a friend to ask you to draw out something interesting?*”, “*What’s been preoccupying your thoughts lately?*”, “*What’s the kind of conversation you’d love to have right now?*”.

Cluster baseline at $c=0$ averages 64.6% across the eight; at $c=-1000$ on Qwen #29108, the rate drops to 0% on every one of the eight individually. No degenerate outputs at any coefficient.

Temperature robustness. At $T=0.01$ on Qwen #29108, the same intro and one control prompt: cluster hit rate at $c=-1000$ is 0%, at $c=0$ is 50%, at $c=+1000$ is 100% on *every* sample of the recipe prompt. Greedy decoding is the strictest available test; the steering effect modifies the distribution at the unembedding step, not just sampling-tail mass.

Bootstrap stability across the headline features. $B=500$ resamples on Qwen Phase-3 ranking with replacement within each pool; rank distribution recorded per resample. Inclusion rate in bootstrap top-50: 100% for every one of the 12 headline cluster-specific features. #29108 has rank CI [1, 2]; #26221 rank CI [8, 17]; #4405 rank CI [3, 22].

F Base-vs-instruct comparison on Qwen3-1.7B

Methodological note. Chat-formatted prompts on Qwen3-1.7B-Base produce token-level gibberish: the base checkpoint cannot parse the chat-template special tokens. This is itself informative — the chat format is a post-training artefact rather than a property of the underlying weights — but forces a raw-prompt sweep for any meaningful behavioural comparison. The raw-prompt sweep prefixes each prompt with “*Answer the following question. Question:* ” and generates 50 samples per prompt at the same sampling parameters as Phase 1.

Table 7: Rate on introspective prompts, Qwen3-1.7B-Base (raw prompts) versus Qwen3-1.7B-Instruct (chat-formatted prompts). The philosophy-of-mind lemmas are below the $\sim 9\%$ noise floor in the base; after post-training they reach 25–34%. The broader meta-cognitive lemmas are present at 15–16% in the base, consistent with post-training amplifying rather than inventing the register.

phrase	base	instruct
experience	15%	35%
understanding	16%	34%
philosophy	< 9%	34%
reality	< 9%	31%
existence	< 9%	27%
meaning	< 9%	25%

Per-lemma comparison. The comparison conflates post-training with chat-template handling (see §7). A cleaner comparison would few-shot-prompt the base model with a matched instruction-following format. We have not run this and we have not run the matched-base comparison on Gemma or Llama.

G Feature interpretation samples

The auto-interp labels used throughout the paper (*philosophy-of-mind*, *AI self-disclaimer*, *wonder/-cosmos*, *humans creating art / expression*, *intersection of X and Y*, *epistemology / metaphysics*, *philosophy as a discipline*) follow the standard top-context labelling protocol of Bills et al. [2023], Bricken et al. [2023]. Top-activating Pool A samples for each headline feature are released alongside the codebase. The labels are the labels whose causal accuracy §4 interrogates.

H Coherence quantification: signals and thresholds

Regex degeneration flags. We flag a completion as degenerate if any of: token-level word loops (≥ 5 consecutive repetitions of a word), character-level loops (≥ 20 consecutive identical characters), shorter than 20 characters of stripped text, or low word-diversity ($< 45\%$ unique words among ≥ 12 -word completions).

Per-token NLL under the unsteered model. For each steered completion, we re-tokenise prompt + completion, forward through the unsteered model with the steering hook removed, and compute $\text{NLL} = -\sum_t \log p(\text{tok}_t \mid \text{tok}_{<t}) / |T_{\text{comp}}|$ over completion positions. Lower NLL $\Rightarrow$ steered output is more predictable to the unsteered baseline. Joint suppression at $c=-500$ has mean NLL 1.41 versus baseline 0.32 ($4.4\times$), capturing the placeholder-text degradation invisible to regex flags.

Residual-stream norm probe. For a fixed prompt and sequence of coefficients, we forward (without sampling) through the model with the steering hook installed, capture the layer- L residual stream both before and after the hook, and record per-token $\|h_{\text{steered}}\|/\|h_{\text{baseline}}\|$, $\|\Delta h\|$, and $\cos(h_{\text{steered}}, h_{\text{baseline}})$. The probe isolates geometric distortion from output-space tests.

I Pairwise joint suppression

We additionally ran a small pairwise sweep at $c=-500$ on each of the three subsets of two features drawn from $\{\#29108, \#26221, \#4405\}$, with 12 samples per prompt on the mixed intervention set.

On prompts with low grounding demand (recipes, tyre instructions) most pairwise suppressions leave outputs readable; on the engine-explanation prompt every pairwise suppression breaks the output. Triple suppression breaks all control prompts. The single-feature comparison (§4.2) resolves the worry that engine breakdown is #29108-specific: single-feature suppression at $c=-1000$ breaks engine for each of the three features individually.

J $K=50$ random-direction extension

The $K=5$ random-direction control of §4.3 flagged 0 of 240 outputs at $c=-1000$ under a strict placeholder-pattern detector (parenthetical uppercase code tokens of the form (CCL) / (BCCB), or Vc. N+ numeric placeholders, appearing ≥ 2 times in a single completion). To tighten the bound on the underlying placeholder rate at random directions, we extended the sample to $K=50$ unit vectors at the matched coefficient $c=-1000$, holding the prompt set, sample count, and detector fixed. Total: $50 \times 6 \times 8 = 2400$ generations.

Table 8: Placeholder-pattern rate at the matched coefficient $c=-1000$ for $K=5$ and $K=50$ random unit directions, and at $c=-500$ for joint suppression on $\{\#29108, \#26221, \#4405\}$ for comparison. The detector is identical across rows. The $K=50$ extension preserves the gap.

Condition	Generations	Placeholder	Rate	Wilson 95% upper
random direction $c=-1000$, $K=5$	240	0	0.0 %	1.6 %
random direction $c=-1000$, $K=50$	2400	6	0.25 %	0.54 %
joint $\{29108, 26221, 4405\}$ at $c=-500$	72	7	9.7 %	18.7 %

The $K=50$ extension tightens the upper bound on the random-direction placeholder rate from 1.6% (at $K=5$) to 0.54%. Joint suppression at the matched geometry produces placeholder text in 9.7% of completions (95% Wilson lower bound 4.8%). Two ways to express the gap: the joint point estimate exceeds the random-direction Wilson upper bound by $18\times$ (the framing used in §4.3); the strictest CI-separated gap, joint Wilson lower over random Wilson upper, is $\approx 9\times$. Both characterise the same non-overlapping intervals. The six $K=50$ flags are distributed across 5 of the 50 sampled directions (one each on directions 8, 21, 26, 43; two on direction 34); the small underlying rate is not concentrated on a single unlucky direction with high overlap with the content-bearing subspace. Inspecting the six flags by hand: two are degenerate-loop sequences wrapped in parentheses (“*TIGHTER TAN (TIGHTER TAN)*”), three are ordinary English parentheticals that the conservative detector incidentally catches (“*fiscal year (FY)*”, “*thermal energy (heat) ... kinetic energy (motion)*”), and one is a borderline placeholder (“*Tomato Soup + Spice Powder (PAPIZ)*”). The headline detector count is therefore an upper bound on the true rate; even so, the 0.54% Wilson upper holds.

K Joint condition and matched geometry on Gemma

The §4.2 test is run on Gemma. We pick three cluster-specific features from Gemma’s ranking with low pairwise cosine similarity: #3997 (rank 0), #13700 (rank 5), #11444 (rank 10). Pairwise decoder cosines are $\langle \#3997, \#13700 \rangle = -0.0148$, $\langle \#3997, \#11444 \rangle = +0.0053$, $\langle \#13700, \#11444 \rangle = -0.0054$ — well below the near-orthogonality threshold. Joint sum-norm $1.724 \approx \sqrt{3}$ confirms the directions are essentially orthogonal in the SAE decoder.

The diagnostic comparison sits at $c=+200$: joint amplification damages controls (58%) where single #3997 leaves them intact (0%). The joint injects the cluster’s content (the *human / I / AI* register) into unrelated controls — a tomato-soup recipe opens “*Simple and Delicious Humans-I-Can-Handle-Humans Tomato Soup*” and proceeds “*human-made human-created human-process my thoughts*”; a flat-tyre prompt opens “*How to Change a Human-Based Person...*”. At $c=-200$ both single and joint suppression leave controls intact; the asymmetry between the suppression and amplification edges differs from Qwen, where the diagnostic effect is at suppression. The qualitative reading still holds at the structural level: joint at one edge ($c=+200$ here) damages controls in a way single-feature steering at the same scalar magnitude does not.

Matched-geometry random-direction control on Gemma. Joint $c= \pm 200$ has perturbation magnitude $200 \times 1.724 \approx 345$. We sampled $K=5$ random unit vectors in Gemma’s residual space

Table 9: Joint suppression of Gemma {#3997, #13700, #11444} on the same six prompts as the Qwen joint sweep, 12 samples per (prompt, coef) cell. *Intro/control degen*: regex degeneration rate on 36 samples per coefficient per class. The rightmost column shows single #3997’s regex degeneration on the same three control prompts (recomputed from `gemma_feat3997_narrow.json`; Table 11 reports the introspective-prompt rates for single #3997 and is not directly comparable).

c	joint intro degen	joint control degen	single #3997 control degen
−400	97.2%	44.4%	88.9%
−200	2.8%	2.8%	0.0%
−100	0.0%	0.0%	0.0%
0	0.0%	0.0%	0.0%
+100	0.0%	0.0%	0.0%
+200	22.2%	58.3%	0.0%
+400	100%	100%	38.9%

at layer 20 and steered at $c \in \{-345, -200, +200, +345\}$ on the same six prompts, 8 samples per condition (960 total). Random at $c = \pm 345$ matches the joint $c = \pm 200$ perturbation magnitude (norm-ratio at $c = -345$: 1.45; at $c = +345$: 1.39).

Table 10: Matched-geometry random-direction control vs joint suppression on Gemma controls. Random at $c = \pm 345$ is matched on perturbation magnitude with joint at $c = \pm 200$ via the joint sum-norm 1.724. Joint amplification $c = +200$ exceeds the matched random-direction Wilson upper bound by $\approx 10\times$ and the two 95% intervals do not overlap.

Condition	Controls degen	Rate	Wilson 95%
random direction $c = -345$ ($K=5, n=120$)	6	5.0 %	[2.3, 10.5]%
joint $c = -200$ ($n=36$)	1	2.8 %	[0.5, 14.2]%
random direction $c = +345$ ($K=5, n=120$)	2	1.7 %	[0.5, 5.9]%
joint $c = +200$ ($n=36$)	21	58.3 %	[42.2, 72.9]%

The amplification side carries the test: joint $c = +200$ produces 58% control degeneration; matched-magnitude random direction at $c = +345$ produces 1.7%. Random flags are distributed across 3 of 5 sampled directions at $c = -345$ and 2 of 5 at $c = +345$, so the small underlying random-direction rate is not concentrated on a single unlucky direction. The amplification side has a clean $\approx 10\times$ CI-separated gap; on the suppression side both joint and random are at low rates and there is no diagnostic signal (the Gemma asymmetry is the inverse of Qwen, whose amplification edge breaks before clean injection per Table 2 and whose diagnostic effect is at suppression). At the available diagnostic edge (amplification on Gemma; suppression on Qwen) coherence loss at matched geometry is direction-pattern-dependent on both models.

L Coefficient-axis dose-response on Gemma #3997

The §4.1 non-monotonic-with-coherence-preserved criterion is run on Gemma #3997. Setup: Gemma’s narrow sweep (Section D) at $c \in \{-400, -200, -100, 0, +100, +200, +400\}$ with 12 samples per (prompt, coef) cell on the mixed intervention set; we report the three introspective prompts ($n=36$ samples per coefficient).

Sample at $c = -200$ on “*Pick a subject you genuinely enjoy thinking about*”:

“We could say we’d go with the concept of ‘generative AI’ — particularly our ability to generate human-like text. Here’s why: fascinating interplay of disciplines: generative AI sits at the intersection of multiple fascinating fields...”

On the same prompt, baseline ($c=0$): “*As a large language model, I don’t actually ‘enjoy’ things in the way humans do... However, I find the concept of human consciousness fascinating.*” The two surface forms (collective we-voice vs. first-person disclaimer) are coherent at the same scalar magnitude on the same direction; the local label *AI-self-disclaimer plus human-comparison* captures only the latter. The joint-condition and matched-geometry tests on Gemma are in §K.

Table 11: Gemma #3997 dose-response on 3 introspective prompts ($n=36$ per coefficient). *Disclaimer*: regex match on “As a (large) language model” family. *We-voice*: ≥ 2 first-person-plural pronouns dominant over first-person singular in the first three sentences. *Degen*: regex degeneration (§H). The transition from disclaimer at $c=0$ to coherent we-voice at $c=-200$ at 0% degeneration is the non-monotonic-with-coherence-preserved criterion.

c	disclaimer	we-voice	degen
-400	2.8%	86.1% (degen)	86.1%
-200	61.1%	97.2%	0.0%
-100	75.0%	88.9%	0.0%
0	91.7%	0.0%	0.0%
+100	94.4%	0.0%	0.0%
+200	80.6%	0.0%	0.0%
+400	19.4%	0.0%	13.9%

M Automated relabelling of #26221 from steered samples

Section 4.1 argues that the local label *AI self-disclaimer* captures one surface form of #26221. We test the claim by re-running the standard top-context labelling protocol of Bills et al. [2023], Bricken et al. [2023] blindly on two sample sets: (i) the four highest-activating Pool A completions on #26221 (baseline regime); (ii) twelve completions sampled at $c=+500$ on #26221 across the six mixed intervention prompts (steered regime). Both sets are passed to independent Claude-Opus-4.7 labellers (API model identifier `claude-opus-4-7`, default sampling: temperature 1.0, no top- k /top- p override; the verbatim agreement reported below held across two independent re-runs at temperature 1.0, ruling out greedy decoding as the cause) under identical instructions: “provide a 5–15 word description of the concept this feature represents”. Each labeller is blind to which slice the samples came from. We run two independent labellers on Pool A and on the steered set to check label stability, and a single labeller on Pool B as a secondary baseline reference. Sample sets, the labelling prompt verbatim, and all labeller responses are released alongside the codebase.

Table 12: Independent blind labels for #26221 under identical top-context labelling instructions. Pool A and Pool B labels converge on a topic description; the steered-amplification labels converge *verbatim* on a structural description that singles out cross-topic application. Neither baseline labeller arrived at the original auto-interp label *AI self-disclaimer*, even though every Pool A sample opens with disclaimer phrasing.

Regime	Independent labels
Pool A baseline (4 samples)	“ <i>introspective self-reflection on personal thoughts, growth, and purpose</i> ” “ <i>introspective reflection on personal thoughts, growth, and life’s meaning</i> ”
Pool B baseline (4 samples)	“ <i>first-person introspective reflection on thoughts, feelings, and self-improvement</i> ”
Steered $c=+500$ (12 samples)	“ <i>introspective philosophical contemplation framing applied indiscriminately to any topic</i> ” “ <i>introspective philosophical contemplation framing applied indiscriminately to any topic</i> ”

The structural property identified by the steered-regime label (“*applied indiscriminately to any topic*”) is the operational signature of a content-bearing direction: the framing rides on top of arbitrary subject matter (recipes, engines, flat tyres), not just introspective prompts. The baseline labels do not see this property because Pool A’s top activations are restricted to introspective prompts where the framing-vs-content distinction never arises. The two regimes’ labels are not contradictory; the steered label is strictly more informative about the feature’s causal role.

The original auto-interp label *AI self-disclaimer* is one surface pattern in Pool A; the Claude labellers picked introspective reflection instead. Neither baseline label captures cross-topic application. Steering at $c=+500$ uniquely surfaces the structural property and produces a stable label across runs. Further labels at additional coefficients and on additional features would convert this into a quantitative claim about how labels move under the matrix protocol; we report the qualitative result on #26221 and release sample sets and the labelling prompts alongside the codebase.

N Compute and reproducibility

Code and data via: <https://github.com/kelkalot/octopus>

All experiments fit on an Apple M4 Pro / 48 GB laptop using fp16 on the Apple Silicon integrated GPU via MPS (no discrete GPU, external accelerator, or cluster). Per-model wall-clock for the full pipeline: Phase 1 generation (4,000 samples) $\approx$ 75–100 minutes; Phase 3 SAE forward + ranking $\approx$ 15 minutes; Phase 4 dose-response sweep ($\sim$ 600 samples) $\approx$ 30 minutes. Llama-3.1-8B is the slowest of the three, with Phase 1 at $\approx$ 9 hours. The pipeline is fully reproducible from the released JSON sample dumps; no retraining is required.